QueryBook

Harvest Network Design Specification

Three sites, three phases: New York headquarters (bunny.net), Boston collector (Cloudflare), Los Angeles collector and backup (OVHcloud)

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Status: design for approval — nothing in this document has been deployed

Applies to: QueryBook Language Builder 1.19, Math, Science & Engineering 1.7, History & News 1.7

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1. Purpose and scope

This specification defines QueryBook's harvest network: one headquarters site that plans, controls, checks and stores every fact, and two remote collection sites that do only what headquarters tells them. It names each site, its location, the recommended products, the information domains and sources it owns, the job and upload protocols, the management interface, security, phases, acceptance tests and estimated costs.

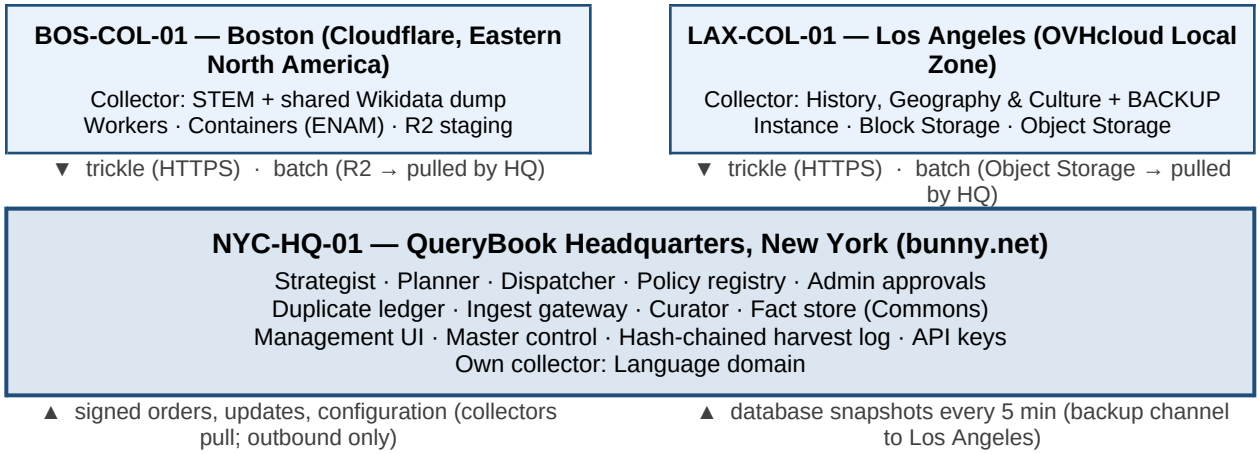
In scope: the three sites, their roles, the collector runtime, job orders, trickle and batch uploads, de-duplication, management and control, phasing.

Out of scope: changes to QueryBook's fact model, the three applications' user features, and any source not yet approved in the policy registry.

1.1 Design goals

- **Speed and efficiency:** bulk downloads first; extract facts next to the source; move only facts; every source harvested in parallel within its own rate allowance.
- **No redundant harvesting:** every source is owned by exactly one site; a central duplicate ledger stops records already held from being fetched or uploaded again.
- **All facts end in New York:** collectors send trickle or batch uploads; only headquarters writes to the fact store.
- **Central control:** collectors hold no QueryBook proprietary software; they are programmed, updated and configured from the headquarters administration screens.
- **Lawful, polite harvesting:** the harvest policy rules (no copyright or licence restrictions, facts only, one key and one rate budget per source) apply to every site.

2. Architecture overview



2.1 Principles

- **Headquarters thinks, collectors obey.** Strategy, plan, policy, de-duplication and curation run only at NYC-HQ-01.
- **Pull, not push, for control.** Collectors open outbound connections only; they ask headquarters for work. No collector exposes a port.
- **One owner per source.** The dispatcher assigns each source to one site; it is reassigned only by the administrator.
- **Facts only.** Collectors return extracted facts (never source wording) with source, licence, URL and a reason, exactly as today's harvesters do.
- **Everything verified at headquarters.** Every upload is re-checked against the policy registry, checksums and the duplicate ledger before any fact is saved.

3. Site summary

Site ID	Name and location	Provider and products	Role	Domains owned	Phase
NYC-HQ-01	QueryBook Headquarters New York, NY (USA)	bunny.net Magic Containers (New York region*), persistent volume, Edge Storage (New York + Los Angeles)	Control, ingest, curation, fact store, management UI; own collector	Language	1
BOS-COL-01	Boston Collector Cloudflare Eastern North America (Boston area*)	Cloudflare Workers (Paid), Containers (region ENAM), R2	Fastest collector: bulk dumps and high-concurrency API work	STEM + shared Wikidata dump	2
LAX-COL-01	Los Angeles Collector and Backup Los Angeles, CA (USA)	OVHcloud Public Cloud Local Zone Los Angeles: instance, Block Storage, Object Storage	Paced API collector; disaster-recovery copy and standby headquarters	History, Geography & Culture	3

* See section 13, items to confirm: bunny.net must offer a New York Magic Containers region; Cloudflare cannot be pinned to Boston itself — Containers are restricted to the Eastern North America region

(ENAM) and Workers are placed near AWS us-east-1, so Cloudflare chooses the exact East Coast data centre.

3.1 Which site pulls the most data fastest

BOS-COL-01 (Cloudflare, Eastern North America) is expected to pull the most data fastest, for four reasons:

- **Proximity:** the largest sources are hosted in the eastern United States (Wikimedia dumps, AWS us-east-1 open-data copies of USDA, Smithsonian and National Archives datasets, NOAA, NASA).
- **Concurrency:** Workers run thousands of small jobs at once; on the Paid plan each invocation may use up to 5 minutes of CPU and 10,000 (configurable higher) outbound requests, and network waiting does not count as CPU time.
- **Bulk capacity:** Containers restricted to ENAM handle large dumps that do not fit a Worker's 128 MB memory.
- **Free egress:** R2 charges nothing for data sent out, so batch bundles reach New York at near-zero transfer cost.

LAX-COL-01 ranks second: steady dedicated instances suit long, paced API harvests, but it is furthest from most sources. **NYC-HQ-01** collects least, so its capacity stays free for checking and storing facts.

4. Site specifications

4.1 NYC-HQ-01 — QueryBook Headquarters, New York (bunny.net)

Item	Specification
Location	New York, NY, USA (bunny.net Magic Containers New York region — to confirm)
Compute	Magic Containers application, 1 instance, 2 vCPU, 4 GB RAM (start at 1 vCPU / 2 GB in Phase 1)
Storage	Persistent volume 100 GB (Commons database, write-ahead log, local backups); Edge Storage zone, Standard tier, regions New York (primary) + Los Angeles (replica) for vault, archives and backups
Network	Anycast IPv4 address; HTTPS only; ingest and fleet endpoints protected by mutual TLS (per-node certificates)
Software	QueryBook core (engine, policy registry, approvals, harvest log, curator, Commons) in headless mode + new fleet services: dispatcher, ingest gateway, duplicate ledger, fleet manager, provider connectors
Holds	All API keys (encrypted, owner-only), the signing key for orders and runner releases, the registry, the fact store
Collects	Language domain (section 5)
Availability	Continuous; Los Angeles standby can take over (section 4.3)

4.2 BOS-COL-01 — Boston Collector (Cloudflare, Eastern North America)

Item	Specification
Location	Cloudflare Eastern North America; Containers constraint regions = ["ENAM"]; Workers placement.region = aws:us-east-1
Products	Workers Paid plan (API fan-out runner); Containers (bulk runner, standard instance type, ENAM only); R2 bucket qb-bos-staging (batch staging, 7-day lifecycle rule)
Runner	Generic open-source job runner, two builds of the same code: a Worker (light jobs) and a container image (bulk jobs)

Item	Specification
Limits to respect	Worker memory 128 MB; Worker CPU up to 5 min per invocation; at most 6 simultaneous outbound connections per invocation
Holds	Nothing persistent except staged bundles in R2 awaiting pickup; API keys only in memory, inside an order
Collects	STEM domain + the shared Wikidata dump (section 5)
Uploads	Trickle: HTTPS to NYC-HQ-01. Batch: bundle to R2, manifest to HQ, HQ pulls (free egress)

4.3 LAX-COL-01 — Los Angeles Collector and Backup (OVHcloud)

Item	Specification
Location	OVHcloud Public Cloud Local Zone, Los Angeles, CA, USA
Compute	1 general-purpose Linux instance, 4 vCPU, 16 GB RAM (runner + standby headquarters)
Storage	Block Storage 500 GB (standby copy of the Commons and archives); Object Storage container qb-lax-staging (batch staging)
Network	Public IPv4 (billed separately in Local Zones); outbound HTTPS; no inbound except an encrypted backup channel from HQ
Runner	Same generic job runner (container image) as Boston
Backup role	Receives HQ database snapshots every 5 minutes and nightly full copies; receives harvest-log archives; can be promoted to headquarters by the administrator
Collects	History, Geography & Culture domain (section 5)
Local Zone limits	No GPU or Windows images; private networks stay inside the zone; SMTP ports blocked

5. Domain and source assignment

Each source has exactly one owner site. Method: **Bulk** = dataset download processed on the collector; **API** = paced requests; **Mixed** = bulk first, then API for changes. Key: whether a free API key is needed (keys stay at HQ and travel only inside encrypted orders). All sources listed are approved in the policy registry; facts only, never wording.

5.1 NYC-HQ-01 — Language

Source (registry key)	Facts	Method	Key	Filter / condition	Priority
Webster 1913 (webster-1913)	English word meanings	Bulk	No	Public domain	1
Wikidata languages (wikidata)	ISO 639-3 codes, writing systems	API between dump refreshes	No	CC0 statements only	1
Tatoeba CC0 sentences (tatoeba-cc0)	Example sentences per language	Bulk	No	Only sentences marked CC0; no translations or audio	2
Common Voice text (common-voice-text)	Sentences in 100+ languages	Bulk	No	Text from the open-source repository only, not the packaged speech datasets	2

Source (registry key)	Facts	Method	Key	Filter / condition	Priority
Gutenberg pre-1931 grammars (gutenberg-pre1931)	Grammar and usage facts	Bulk	No	US public domain; licence wrapper stripped	3

5.2 BOS-COL-01 — STEM and the shared Wikidata dump

Source (registry key)	Facts	Method	Key	Filter / condition	Priority
Wikidata full dump (wikidata)	All domains: countries, people, places, compounds, species, languages, works	Bulk (Container)	No	CC0 statements; one pass, HQ routes facts to domains	1
USDA FoodData Central (usda-fooddata-central)	Nutrients per 100 g	Bulk	api.data.gov	CC0	1
NCBI Taxonomy (ncbi-taxonomy)	Species, common names, groups	Bulk (taxdump)	No	NCBI-created data only	1
Smithsonian science (smithsonian-open-access)	Specimens and instruments: date, place, topic	Bulk (AWS open data)	api.data.gov	Records marked CC0 only	2
PubChem computed (pubchem-computed)	Formula, molecular weight, identifiers	Bulk	No	NCBI-computed values only, no depositor text	2
NASA (nasa)	Space and earth-science facts	API	Optional	Exclude logos, third-party and non-US agency items	3
NOAA (noaa)	Climate and ocean data points	Bulk	No	Exclude annotated products	3
NIST post-quantum standards (nist-pqc-fips)	Algorithm names, parameter sets	Bulk (PDF metadata)	No	US federal	3
arXiv CC0 papers (arxiv-cc0)	Paper metadata and stated results	API (OAI-PMH)	No	Papers licensed CC0 only	3
NIST DADS (nist-dads)	Algorithm and data-structure definitions	Bulk	No	Skip CRC Press entries	3

5.3 LAX-COL-01 — History, Geography & Culture

Source (registry key)	Facts	Method	Key	Filter / condition	Priority
Natural Earth (natural-earth)	Cities, regions, coordinates, capitals	Bulk	No	Public domain	1
USGS GNIS (usgs-gnis)	US place and feature names, coordinates	Bulk	No	US federal	1
Census Gazetteer (census-gazetteer)	Places, counties, areas	Bulk	No	Facts only; no 'TIGER' branding	2
FRUS (frus)	Diplomatic volumes, years covered	Bulk (XML)	No	US federal	1
The Met (met-open-access)	Artworks: artist, date, medium	Bulk (CSV), API for changes	No	CC0 metadata	1

Source (registry key)	Facts	Method	Key	Filter / condition	Priority
National Archives (nara-catalog)	Records, dates, series	Mixed	nara	Use restriction 'Unrestricted' only	1
DPLA (dpla-metadata)	US library and museum records	Mixed	dpla	Metadata only (CC0)	2
Europeana (europeana-metadata)	European heritage records	Mixed	europeana	Metadata only (CC0)	2
Rijksmuseum (rijksmuseum-open-data)	Dutch and European art and history	Bulk	No	Per-object licence field must be CC0	3
Smithsonian history (smithsonian-open-access)	Objects: date, place, type	Bulk (AWS open data)	api.data.gov	Records marked CC0 only	2
Chronicling America pre-1931 (chronicling-america-pre1931)	Dated events (kept as 'reported')	Bulk	No	Titles over 95 years old only	3
HABS/HAER/HALS (habs-haer-hals)	Buildings, bridges, engineering works	Bulk	No	Skip items crediting a third party	3
World Factbook archive (world-factbook-2020-archive)	Country facts 'as of' edition year	Bulk	No	Never stored as current	3

Held back (not assigned): FSI language courses and FHWA reports remain pending under the administrator's condition; Wikimedia, NCBI, the Met and other APIs are never harvested from two sites at once.

6. Job orders (headquarters → collectors)

A collector does nothing without a valid order. Orders are leased: the collector must renew the lease while it works; if headquarters stops renewing (Stop all, node paused, source paused), the collector stops within seconds.

6.1 Order fields

Field	Meaning
order_id, issued_at, expires_at	Unique ID; validity window (default 15 minutes, renewable)
node_id	The single site allowed to run it
source_key	Registry key; must be approved at issue time
allow_hosts	The only hosts the collector may contact for this order
actions	Ordered list of runner actions (section 7) with parameters
recipe	Declarative extraction map (JSON paths, XML paths or CSV columns → subject/predicate/object, evidence, reason)
rate	Requests per second and simultaneous connections allowed (the source's single budget)
known_digest	Compact summary (Bloom filter) of record fingerprints HQ already holds, for early duplicate dropping
upload	trickle or batch, target, batch size, compression
secret (optional)	Encrypted API key for keyed sources; decrypted in memory only, never written

Field	Meaning
signature	Ed25519 signature by HQ's order-signing key over all fields

6.2 Example

```
{
  "order_id":    "ord-2026-10-19-000412",
  "node_id":     "BOS-COL-01",
  "source_key":  "usda-fooddata-central",
  "issued_at":   "2026-10-19T16:00:00Z",
  "expires_at":  "2026-10-19T16:15:00Z",
  "allow_hosts": ["fdc.nal.usda.gov"],
  "actions": [
    {"download": {"url": "https://fdc.nal.usda.gov/.../foundation_food_csv.zip", "resume":
true}},
    {"extract":  {"recipe": "fdc-foundation-v1"}},
    {"filter":   {"known_digest": true}},
    {"upload":   {"mode": "batch", "target": "r2://qb-bos-staging/ord-...-000412/"}}
  ],
  "rate":      {"rps": 2, "connections": 2},
  "signature": "ed25519:..."
}
```

7. Collector runtime (identical at every collector)

The runner is generic software, published as open source, so no collector holds QueryBook proprietary software. QueryBook's strategy, registry, duplicate ledger, fact model and curation remain only at headquarters. Recipes arrive with each order and are discarded after use.

7.1 Actions

Action	Does	Notes
FETCH	One HTTP GET	Conditional requests (If-None-Match / If-Modified-Since); only allow_hosts
PAGE	Follow pagination	Cursor, offset or page rules given in the order
DOWNLOAD	Bulk file	Resumable (HTTP ranges), checksum verified when the source publishes one
EXTRACT	Apply the order's recipe	JSON, XML, CSV; streaming, constant memory
FILTER	Drop records HQ already holds	Uses known_digest; duplicates never uploaded
UPLOAD	Trickle or batch	Section 8
HEARTBEAT / RENEW	Report health; keep the lease	Every 10 seconds while working
REPORT	Counts, timings, errors	Feeds the learning agent and the dashboards

7.2 Rules the runner enforces

- Refuses unsigned, expired or wrongly addressed orders, and hosts not in allow_hosts.
- Never exceeds the order's rate; backs off on HTTP 429/503 and reports it.
- Keeps no database, source list or key; scratch space is wiped after each order.
- Accepts only signed runner releases and signed configuration from headquarters.
- Opens no inbound ports; authenticates to headquarters with its own certificate.

7.3 Programming, updating and configuring from headquarters

Site	Deploy / update channel	Configured by HQ
NYC-HQ-01	Local container (same image)	Concurrency, rate, features
BOS-COL-01	Cloudflare API: Worker script versions and container image tags	Placement region, concurrency, rate, R2 lifecycle
LAX-COL-01	OVHcloud API + signed container image pull	Concurrency, rate, backup schedule

8. Uploads to headquarters

	Trickle	Batch
Use for	API harvesting, steady small output	Bulk dumps, large catalogue runs
Path	Collector → HTTPS POST to NYC-HQ-01 /ingest/trickle	Collector → own staging (R2 or OVH Object Storage) → manifest to /ingest/batch/announce → HQ pulls → verifies → imports → deletes staged copy
Size	Up to 5 MB compressed per request (JSON lines)	Any size; files split at 256 MB
Integrity	Batch ID (idempotent) + SHA-256 of body	Manifest lists every file with SHA-256 and record counts
Failure handling	Retry with exponential backoff; HQ deduplicates by batch ID	HQ retries the pull; partial imports roll back whole
Back-pressure	HQ may answer 'slow down'; the runner obeys	HQ pulls when it has capacity

On arrival, every fact passes the same checks as today: policy registry (source approved, licence class), facts-only rule (wording removed), duplicate ledger, quality checks, cross-checks between the applications, then the curator saves it and writes the hash-chained harvest log entry with the node ID and order ID.

9. Speed, efficiency and de-duplication

- **Bulk first:** the planner chooses a dataset download over an API whenever the source offers one.
- **Extract at the source:** bulk files are processed on the collector; only facts travel to New York (typically a small fraction of the raw size).
- **Changes only:** conditional requests, change feeds (OAI-PMH 'from' dates, Wikidata incremental dumps) and ETags avoid re-downloading unchanged data.
- **Duplicate ledger at HQ:** fingerprints of every record and fact held; summarised into each order so collectors drop duplicates before uploading.
- **One budget per source:** the dispatcher holds each source's rate allowance centrally; speed comes from many sources in parallel, never from hammering one.
- **Placement:** the dispatcher sends work to the owner site; the administrator can reassign a source in one step.
- **Learning under approval:** the existing learning agent proposes batch sizes and schedules from collector reports; nothing changes until the administrator approves.

10. Headquarters management interface

Extends the existing Harvesting & shared facts page. All views are available only at headquarters, behind the administrator passphrase for changes.

View	Shows	Controls
Master control	Overall state; per site, per source	Start / stop all, per site, per source, selected
Fleet	Each node: provider, region, runner version, health, current orders, data moved, cost	Enrol, pause, drain, restart, revoke certificate, update runner
Domains & sources	Owner site, progress, rate budget used, duplicates skipped, last change pulled	Reassign owner, pause source, adjust budget (with approval)
Job board	Queued, leased, running, done, failed orders	Re-queue, cancel, prioritise, force to a node
Uploads	Trickle rate, batch manifests, checksum results, quarantine	Approve or reject quarantined uploads
Costs & budgets	Compute and transfer per provider per day	Monthly caps that pause a site when reached
Audit	Orders, uploads and admin actions in the hash-chained log	Verify chain, export, archive
Honesty checks	Results of HQ's random re-fetch of collector work	Quarantine a node automatically on mismatch

11. Security

- **Identity:** each collector enrols once with a one-time code and receives its own certificate; mutual TLS on every connection; revocation from the Fleet view.
- **Signing:** orders, runner releases and configuration are signed by HQ (Ed25519); collectors refuse anything else. Plan migration of signing to NIST post-quantum signatures (FIPS 204) when library support is mature.
- **Secrets:** API keys and signing keys live only at NYC-HQ-01 (owner-only files, never in the database, backups or logs). Keys reach a collector only inside an encrypted order, in memory only.
- **Network:** collectors expose no inbound ports; HQ exposes only the fleet and ingest endpoints (mutual TLS) and the administration interface (restricted to the administrator's addresses or a private tunnel).
- **Integrity:** checksums on every upload; hash-chained harvest log; random re-fetch audits of collector output.
- **Policy:** the registry check runs at order issue and again at import; a source blocked by the administrator stops immediately everywhere.

12. Phases, deliverables and acceptance

Phase	Duration	Deliverables	Acceptance tests
1 — NYC-HQ-01	3–4 weeks	Headless HQ on bunny.net; dispatcher; signed orders; ingest gateway (trickle + batch); duplicate ledger; fleet, job board and upload views; generic runner run locally at HQ for the Language domain	Language sources harvested end to end through orders; Stop all halts within 10 s; duplicate upload rejected; tampered batch rejected; chain verifies
2 — BOS-	2 weeks	Runner as Worker + Container (ENAM); R2	Wikidata pass completes and facts

Phase	Duration	Deliverables	Acceptance tests
COL-01		staging; Cloudflare deploy connector; first full Wikidata dump pass; STEM sources	route to all three domains; no source harvested from two sites; R2 batch pulled and verified; runner update deployed from HQ
3 — LAX-COL-01	2 weeks	Runner on OVHcloud Local Zone; Object Storage staging; OVHcloud connector; backup channel; standby HQ	History & Geography sources harvested; snapshots every 5 minutes; standby promoted in a drill within 30 minutes with no fact lost

Total build: about **7–8 weeks**. Each phase ends with the administrator's sign-off before the next begins.

13. Estimated monthly cost and items to confirm

Site	Items	Estimate (USD/month)
NYC-HQ-01 (bunny.net)	Magic Containers 1–2 vCPU, 2–4 GB, 100 GB volume, IPv4; Edge Storage 2 regions	30–55
BOS-COL-01 (Cloudflare)	Workers Paid base $5 + usage; Containers usage during bulk runs; R2 storage and operations (egress free)	15–40
LAX-COL-01 (OVHcloud)	4 vCPU / 16 GB instance, 500 GB Block Storage, Object Storage, IPv4	40–70
Total		about 85–165

Estimates from providers' published prices (October 2026); confirm before ordering.

13.1 Items to confirm before Phase 1

1. bunny.net offers a New York region for Magic Containers (otherwise use the nearest US East region).
2. Cloudflare: exact Containers instance sizes, disk and pricing; confirm ENAM capacity for multi-hour bulk jobs.
3. OVHcloud: Local Zone Los Angeles instance types, and whether outbound traffic is included.
4. Each provider's acceptable-use terms permit automated data collection from public sources.
5. Free API keys obtained (api.data.gov, DPLA, National Archives, Europeana) and entered at HQ only.

14. References

- bunny.net Magic Containers — https://bunny.net/docs/magic-containers
- bunny.net Magic Containers pricing — https://bunny.net/docs/magic-containers/pricing.md
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- QueryBook harvest policy registry and engine — querybook/apps/README.md (this repository)